Clinical Term Mapping Engine: ICD-10-CM & SNOMED CT

HiLabs Hackathon 2026 · technical write-up · demo video: (link to be added) · repo: dhisanthks/hilabs (private)

All numbers are 5-fold cross-validated on the training data. The evaluation set contains **diagnosis / condition terms only**, so every SNOMED CT number below is on diagnosis terms (all ICD rows are diagnoses).

5-fold CV (diagnosis terms)	ICD-10-CM	SNOMED CT
Exact match	**71.1%**	**67.4%**
Codes never seen in training (long tail)	50.1%	57.4%
Rare or never-seen codes (≤ 2 training examples)	61.5%	63.3%
Partial credit: ICD 3-character category / SNOMED semantic tag	81.6%	84.4%
Exact or near-synonym (name similarity ≥ 90)	74.4%	73.9%
Most-confident 30% of terms (abstain on the rest)	96.4%	95.9%

1. Approach — problem and what the data told us

Input: an entity extracted by CodEx plus its source chunk; output: one ICD-10-CM code and one SNOMED CT concept from the supplied dictionaries (98,222 ICD codes; 382,124 SNOMED concepts, 984,585 descriptions). 9,646 labelled rows (2,306 ICD, 7,340 SNOMED of which 2,700 diagnosis terms) shaped every decision:

• **Every entity string occurs once** — lookup is useless; the engine must generalize to unseen wording.

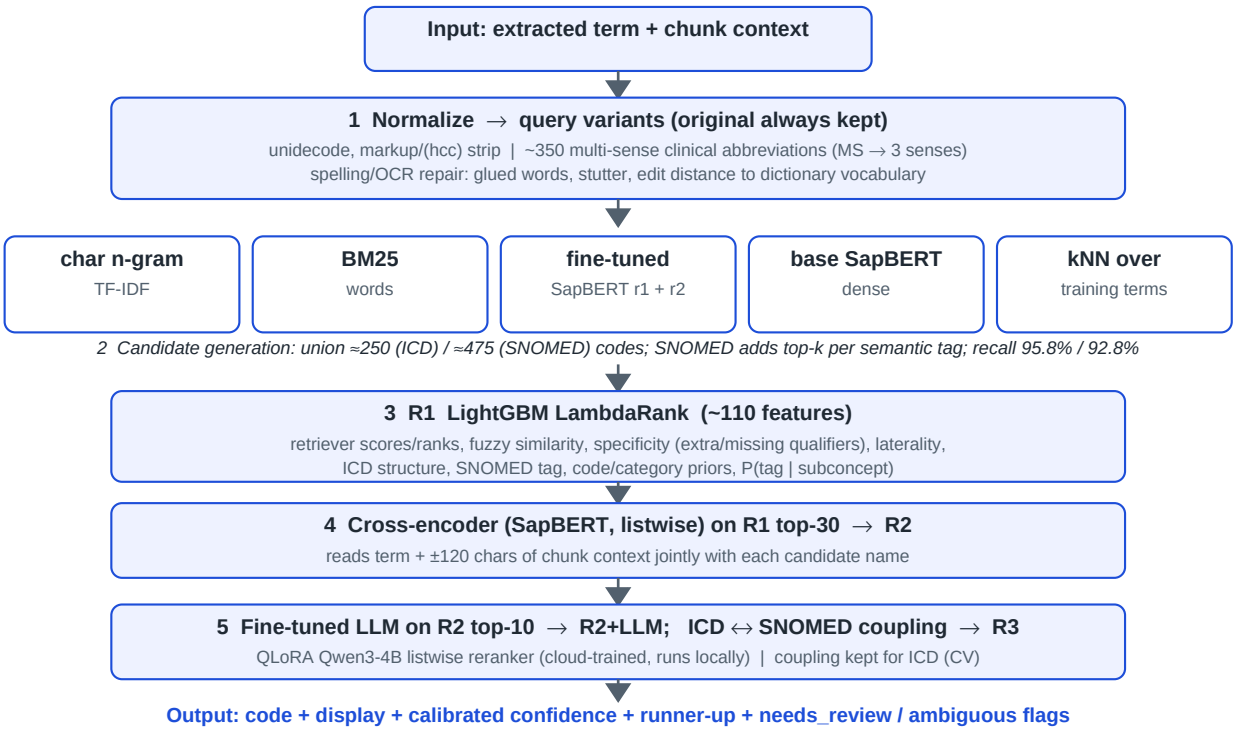
• **Local conventions dominate** — only 6% (ICD) / 15% (SNOMED) of terms match a dictionary string exactly (*can not squat* → R26.2 Difficulty in walking; *rashes* → Eruption).

• **Long tail** — 24% (ICD) / 46% (SNOMED diagnosis) of held-out terms have a gold code never seen in training folds.

• **Label ambiguity** — see §6: the gold standard's own consistency is 93.7% / 84.6%.

Protocol. 5-fold CV grouped by source chunk; every learned component (encoders, cross-encoder, LLM, kNN memory, priors, rankers) is re-trained inside each fold, so every number is on terms the system never saw.

2. Architecture



3. How accuracy was built up (each step measured, 5-fold CV)

Step	ICD	SNOMED (diagnosis)
char TF-IDF + nearest-training-term baseline	41.2	n/m*
+ normalization + R1 LambdaRank over lexical retrievers	55.9	53.4
+ SapBERT dense retrieval and kNN (best of 7 encoders benchmarked)	63.1	n/m*
+ per-fold fine-tuned SapBERT + tag-stratified SNOMED retrieval	66.7	62.0
+ cross-encoder with chunk context (R2)	67.5	62.3
+ ICD↔SNOMED coupling (R3)	67.7	62.2
+ round-2 fine-tuning (iterative hard negatives), ensembled with round-1 encoder	68.3	62.7
+ cross-encoder v2 (negatives from the final ranker, 2 epochs)	68.4	63.2

Step	ICD	SNOMED (diagnosis)
+ chunk-context features (rankers see the chart context)	68.9	63.6
+ fine-tuned LLM reranker (R2+LLM) — final	**71.1**	**67.4**

* n/m: these two early stages were scored only on all SNOMED rows, not on the diagnosis subset.

4. How the models are trained

• **Fine-tuned SapBERT bi-encoders (retrieval + kNN features, feed R1).** Start from SapBERT (PubMedBERT trained on UMLS synonyms). Contrastive fine-tuning on (term → gold code description) pairs with MultipleNegativesRanking loss: every other pair in the batch is a negative, plus explicit hard negatives; 2 epochs, lr 2e-5, batch 64, re-trained per fold. *Round 1* uses the lexically closest wrong code as the hard negative. *Round 2* uses the 3 wrong codes the round-1 model itself ranked highest (iterative hard negatives) — typically sibling codes differing only in specificity (E11.40 vs E11.42, unspecified vs specific), so the encoder learns the coders' specificity and wording conventions. Both rounds are kept as complementary feature sets (ensemble).

• **R1 / R2 / R2+LLM / R3 — LightGBM LambdaRank.** Each term's candidates form one ranking group (label 1 = gold). LambdaRank grows trees that push the gold above wrong candidates, weighting each pair by its effect on top-1 (NDCG). This is how retriever weights are learned: the 'weight' of a retriever is not a fixed number — trees split on each retriever's score and rank *in combination* with other features (e.g. trust nearest-training-term votes when several neighbours agree; distrust dense similarity when the candidate adds an unsupported qualifier), and the criterion is purely which combination puts the gold first on training folds. A fixed quality-weighted fusion was tested and lost (38.0% vs 51.5% top-1). Each ranker is 5-seed bagged (70% rows / features per model).

• **Cross-encoder (R2).** SapBERT as a pair classifier reading [term | ±120 characters of chunk context] with each candidate name; listwise softmax loss over the gold + 9 hard negatives taken from R1's own top-30; 2 epochs, per fold.

• **Fine-tuned LLM reranker (R2+LLM).** Qwen3-4B-Instruct-2507 (Apache-2.0) loaded in 4-bit and fine-tuned with QLoRA (LoRA rank 16, α 32, on all attention and MLP projections; ~0.8% of weights trainable). Prompt: coding system, entity type, term, chart context and R2's top-10 candidates lettered A–J; the training loss is cross-entropy over the 10 letter logits at the answer position (listwise, no text generation); candidate order is shuffled during training so it cannot learn position. 1 epoch, lr 1e-4, honest 5-fold on Kaggle 2× T4 + a final adapter trained on all data. At inference one forward pass gives a log-probability per candidate (deterministic) → features llm_lp / llm_rank / llm_gap for the R2+LLM ranker. Gain: +2.2 ICD, +3.8 SNOMED diagnosis.

5. Design decisions (and the stage each one lives in)

• **[Retrieval] Many cheap retrievers, union of candidates** (char n-gram, BM25, fine-tuned + base SapBERT, kNN over training terms; SNOMED also top-k inside each semantic tag) → 96% of gold codes in the pool.

• **[Normalization] Variants, never replacements** — abbreviation senses (MS → 3 senses), spelling/OCR repair, glued words.

• **[R1] Retriever weighting learned, not hand-set** — LambdaRank over ~110 per-candidate features (see §4).

• **[R1] Specificity calibration** — handled by the round-2 fine-tuned SapBERT (trained against sibling codes) plus explicit R1 features: qualifier words a candidate adds (over-coding) or misses (under-coding), unspecified / other / leaf flags. Specificity errors are balanced (46 over- vs 60 under-coded), i.e. no systematic bias.

• **[R1] Context** — chunk-context type classifier and chunk-style flags let the ranker use the chart context.

• **[R2] Cross-encoder** reads term + context jointly with each candidate — resolves abbreviation collisions from context.

• **[R2+LLM] LLM where judgement is needed** — only on R2's top-10, as features, not as a hard decision.

• **[R3] Independent prediction, then coupling** — a learned crosswalk recovers only 37% of ICD codes even from the gold SNOMED code, so each system is predicted on its own; the other system's top-5 enters R3 as evidence. Kept for ICD, not for SNOMED (final stage R2+LLM).

• **[Deployment] Deterministic and private** — fixed seeds, no sampling; everything runs locally on a 4 GB GPU: ~0.2 s/term without the LLM, ~0.85 s/term with it (two-phase inference); dictionary files are fingerprinted for version drift.

6. Error analysis, label ambiguity and flagging

Failure mode (share of errors)	ICD (667)	SNOMED diagnosis (879)
gold never retrieved	14%	13%
right ICD category, wrong specificity	34%	—
right chapter, wrong category / same SNOMED tag, wrong concept	24%	48%
wrong chapter / wrong SNOMED semantic tag (mostly disorder ↔ finding)	28%	39%
…of which the predicted name is a near-synonym of the gold (≥ 90)	10%	20%

• **Label ambiguity.** Near-identical training terms share the majority gold code only **93.7% (ICD) / 84.6% (SNOMED)** of the time (*ble edema* → R60.9 vs *mild ble edema* → R60.0; *phq-9* vs *phq 9* → different concepts) — matching the ~0.85 inter-annotator agreement of the SNOMED CT Entity Linking Challenge. 20% of SNOMED diagnosis errors are near-synonyms (*Hyperhidrosis* vs *Diaphoresis*); crediting them gives 73.9%. Twin-preference features and label cleaning could not learn the coders' tie-breaking (±0.2).

• **Hard slices.** ICD chapter Z (history/status) 47.5%, T (injury) 51.4%; SNOMED diagnosis terms whose gold is a rare type (situation 18.5%, procedure 20.6%); context-dependent abbreviations still fail on ICD (*MS* in an echo report → multiple sclerosis; SNOMED correctly gives mitral valve stenosis).

• **Flagging mechanism.** confidence = softmax probability of the chosen code over the final ranker's scores. *needs_review* = confidence < 0.5 → flags 30% of SNOMED diagnosis terms, which are only 32.8% correct, while the unflagged 70% are 82.4% correct. *ambiguous* = runner-up probability ≥ half the winner's, or runner-up name a near-synonym (≥ 90); every row also carries the runner-up code (alt_code). Confidence > 0.9 is right 96.1% (SNOMED dx); accuracy at 70% / 50% / 30% coverage: SNOMED dx 82.4 / 90.1 / 95.9%, ICD 85.4 / 92.3 /

96.4%.

7. Core requirement: the seven hard cases

Failure mode	How the engine handles it	Evidence
1 Nearest string is wrong	~110 signals beyond string similarity; specificity features penalize extra qualifiers	HTN → I10; pulmonary hypertension → I27.20
2 Abbreviation collision	all senses retrieved; cross-encoder + LLM read the chunk context; low confidence → review	MS (echo) → mitral valve stenosis
3 Specificity calibration	round-2 fine-tuned SapBERT (sibling negatives) + R1 add/miss-qualifier features	diabetes → E11.9; t2dm with diabetic neuropathy → E11.40
4 Granularity mismatch	both systems predicted independently; coupling only as R3 evidence	crosswalk alone: 37%
5 Long tail	dictionary-wide retrieval, fine-tuned encoders, LLM knowledge	never-seen codes 50.1% / 57.4%
6 Upstream noise	markup strip, glued-word split, stutter cut, spelling repair (as variants)	chronickidney disease stag 3 → N18.30
7 Version drift	dictionary fingerprint → all indexes rebuilt; new codes retrievable at once	README §Version drift

8. Answers to the brief's open questions

• **Partial credit?** Exact match as headline; report hierarchical credit (ICD category 81.6%, SNOMED tag 84.4%).

• **Ambiguous gold?** Accept sets of valid codes (20% of SNOMED diagnosis errors are near-synonyms); rows carry alt_code.

• **Inter-annotator agreement?** Publish it as the ceiling: 93.7% / 84.6% measured; 95% exact match exceeds it.

• **Per-category floors?** Most ICD chapters 67–82%; a ~50% floor would flag chapter Z (47.5%) and rare SNOMED gold types.

9. What we tried and abandoned

Idea	Result	Decision
Zero-shot local LLM judge (Qwen3-4B, few-shot)	CV blend +0.25	replaced by fine-tuned LLM
2nd LLM: SNOMED-diagnosis specialist over the top-20	stacked 67.44% = without it	dropped (redundant)
Cross-encoder v3 with the entity type in the query	ICD +0.6, SNOMED –0.7	dropped
Quality-weighted retriever fusion (RRF, MRR weights)	38.0% vs 51.5% top-1; as feature ±0.2	dropped
Concept-type classifier / two-level type ranker	type accuracy = R1's own	dropped
Context-kNN; twin / majority-convention features; cleaned labels	±0.2 (noise)	dropped
Diagnosis-only specialized ranker	62.9% vs 63.2% (all-rows ranker)	dropped
Longer encoder training (4 epochs); bigger LightGBM	±0 / ±1	harder negatives instead

10. Challenges & learnings

• **The data sets the ceiling:** measuring label consistency early would have focused effort sooner.

• **New signals beat new features:** fine-tuning, cross-encoder and the fine-tuned LLM gave the gains; eight feature ideas gave ≈ 0.

• **Only trust full 5-fold numbers:** a one-fold check overstated the cross-encoder (+3.5 vs +0.6).

• **A 4 GB GPU shaped the engineering:** two disk-full incidents (page-file growth), memory-mapped caches, atomic saves, two-phase inference (5.7 → 0.85 s/term).

11. Innovations

Tag-stratified SNOMED retrieval; iterative hard-negative fine-tuning and a two-round encoder ensemble; explicit specificity features; chunk-context features for the rankers; a cross-encoder trained on the final ranker's own confusions; a listwise QLoRA-tuned LLM reranker read by letter log-probabilities (deterministic, cloud-trained, runs locally); calibrated confidence with review/ambiguity flags and runner-up codes; measured label-consistency ceiling and tolerant metrics; dictionary fingerprinting for version drift.

12. Future improvements

• **More HiLabs-labelled diagnosis data** — 46% of SNOMED diagnosis terms have a gold code never seen in training.

• **Stronger LLM** — 7–8B model or more epochs on larger GPUs (the 4B model gave the largest late gain; a second 4B model over the top-20 was redundant, so capacity is the next lever); apply it only to flagged terms to keep cost low.

• **UMLS licence** → SNOMED hierarchy and official map: merge disorder/finding twins, hierarchical credit, better coupling.

• **Synthetic chart-style training terms** from public terminology names (local LLM, no patient data) for unseen codes.

• **Evaluation & workflow** — set-based scoring for ambiguous terms, published coding conventions, and active learning: route the 30% flagged terms to coders and feed their decisions back into training.

13. External resources

SapBERT (cambridgeltl/SapBERT-from-PubMedBERT-fulltext, Apache-2.0) — bi-encoders and cross-encoder. Qwen3-4B-Instruct-2507 (Apache-2.0; unsloth 4-bit build) fine-tuned with QLoRA (PEFT, bitsandbytes) on Kaggle 2× T4 GPUs. Benchmarked only: BioLORD-2023, gte-base, MedCPT, BGE-base-en-v1.5, MedEmbed-base-v0.1, e5-base-v2. ICD-10-CM and SNOMED CT US dictionaries as supplied; hand-curated clinical abbreviation list; PyTorch, sentence-transformers, transformers, LightGBM, scikit-learn, RapidFuzz. Related work: DrivenData SNOMED CT Entity Linking Challenge, n2c2 2019 Track 3, xMEN, MedCAT (docs/RESEARCH.md).

Reproduce: see README — `python run.py predict --input eval.csv --output predictions.csv --assume-subconcept DiagnosisCode` with the released models.

Appendix A — Long-tail generalization (accuracy by gold-code frequency in the training folds)

Gold code seen in training	ICD share	ICD accuracy	SNOMED dx share	SNOMED dx accuracy
absent (never)	23.8%	50.1%	46.2%	57.4%
rare (1–2 times)	24.4%	72.6%	29.8%	72.4%
3–5 times	21.9%	80.0%	16.4%	80.8%
6–20 times	29.2%	80.0%	7.6%	80.9%
rare or absent (≤ 2)	48%	61.5%	76%	63.3%

Appendix B — Accuracy by category (final model)

ICD chapter	n	accuracy	SNOMED diagnosis gold type	n	accuracy
N / E / B	68 / 131 / 30	82.4 / 80.2 / 80.0%	disorder	1,276	76.1%
C / G / K	50 / 79 / 102	78.0 / 77.2 / 76.5%	finding	1,035	68.5%
J / I / R	143 / 292 / 690	74.1 / 74.0 / 72.0%	situation	65	18.5%
M / D / H / F	293 / 28 / 62 / 46	67.9 / 67.9 / 67.7 / 67.4%	procedure	34	20.6%
S / L / T / Z	50 / 86 / 37 / 80	66.0 / 62.8 / 51.4 / 47.5%			

Appendix C — Zero-shot encoder benchmark (retrieval recall over all rows, kNN@1)

Encoder	ICD R@1	ICD R@100	ICD kNN@1	SNOMED R@1	SNOMED R@100	SNOMED kNN@1
SapBERT (chosen)	32.8	76.5	48.5	38.6	79.2	26.7
BioLORD-2023	33.6	81.1	46.5	36.7	75.4	25.3
gte-base	28.3	75.3	47.1	38.0	80.2	25.8
MedCPT	25.3	79.3	47.2	35.2	77.6	25.0